Mini Case Study on Machine Learning Research

Application of Machine Learning in Species Distribution Modelling

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1. Introduction and Choice of Case Study

This report is a critical review of the paper by Iyer, Banerjee and Bhowmick [1], which models the potential distribution of the invasive plant Mikania micrantha in India using lasso and ridge regularised logistic regression. We chose it because Species Distribution Modelling (SDM) is a compact, real-world setting in which almost every stage of the ML pipeline matters: noisy presence-only data, artificially generated negative examples, highly correlated predictors, and a genuine tension between predictive accuracy and interpretability. The paper is also a clean example of adapting a classical model (logistic regression) to the needs of a domain rather than simply fitting a new algorithm to new data.

To test the paper's ideas rather than only describe them, we also built a parallel framework for a second invasive species, Lantana camara, using PCA with Decision Tree and K-Nearest Neighbours (KNN) classifiers. Throughout the report, material reported by the original authors is labelled as the reference paper's, and our own implementation and judgements are labelled as ours.

2. Research Objectives

2.1 Reference paper

The paper has a primary and a secondary objective. The primary objective is to predict the potential geographic distribution of M. micrantha in India and to identify the climatic conditions that facilitate its rapid spread. The secondary, methodological objective is to demonstrate that statistical regularisation (lasso and ridge) can resolve the multicollinearity that is common in environmental datasets, while also identifying the climatic variables that matter [1].

2.2 Our extension (Lantana camara)

The reference study used only climatic variables. Our objectives were (i) to model a different invasive species, L. camara, in India; (ii) to add anthropogenic and hydrological proximity variables (distance to railways, roads, urban areas, rivers and lakes) alongside the 19 bioclimatic variables; and (iii) to handle multicollinearity with Principal Component Analysis (PCA) instead of built-in regularisation, followed by non-linear classifiers.

3. Machine Learning Approach

The task is binary classification: given the environmental conditions at a location, predict whether the species is present (1) or the point is a background point (0). The two studies differ in how they treat the model and the multicollinearity problem (Table 1).

Table 1. Comparison of the reference paper's approach and our extension

Aspect	Reference paper [1]	Our extension
Type of ML contribution	Existing model adapted: logistic regression with l1 (lasso) and l2 (ridge) penalties	Existing models re-arranged: PCA as a separate step, then standard Decision Tree and KNN
Handling of multicollinearity	Built into the model: penalties shrink the coefficients of correlated variables; lasso drops irrelevant ones	Separate preprocessing: standardise, then PCA to uncorrelated components
Model family	Linear (parametric)	Non-linear (tree-based and instance-based)
Predictors	Climate only (WorldClim BIO1-BIO19)	19 climate + 5 proximity variables (24 in total)
Target species	Mikania micrantha	Lantana camara

The rationale in the reference paper is parsimony: lasso shrinks some coefficients exactly to zero, so the final model contains a small set of interpretable climatic indicators. Our rationale was different. Real species-environment relationships are rarely linear, so we wanted classifiers able to capture threshold and interaction effects, and PCA gave us a way to remove correlation without discarding any variable.

4. Data and Machine Learning Pipeline

4.1 Reference paper pipeline [1]

Occurrence records of M. micrantha came from the Global Biodiversity Information Facility (GBIF) [2] and regional herbaria. Nineteen bioclimatic variables were taken from WorldClim [3]. Random background (pseudo-absence) points, equal in number to the occurrence points, were generated over different geographic extents. Logistic, ridge-logistic and lasso-logistic models were then fitted and compared.

4.2 Our pipeline

Figure 1 summarises the pipeline we implemented in Python (pandas, GeoPandas, rasterio and scikit-learn [6]-[8]); the corresponding code is given in the Appendix.

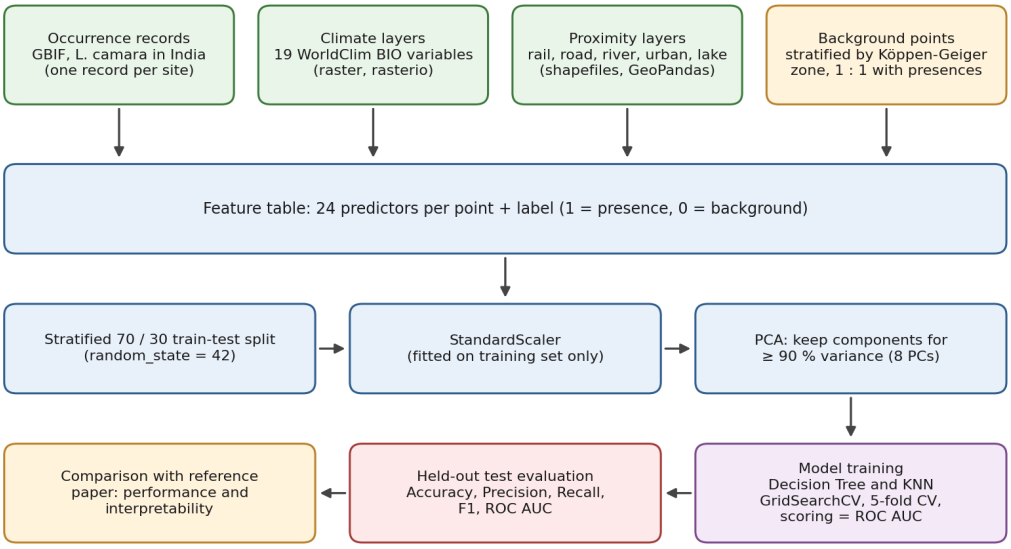


Figure 1. Machine learning pipeline of our Lantana camara extension (diagram prepared by the authors).

- **Occurrence data.** Geo-referenced L. camara records (decimalLongitude, decimalLatitude) were taken from GBIF [2], with one record retained per site to limit spatial autocorrelation. This gave 3,464 presence points within India.
- **Predictors.** The 19 WorldClim bioclimatic variables [3] were read from raster files at each point. Distances to railways, roads, rivers, urban areas and lakes were computed from shapefiles using GeoPandas spatial joins [6], [7].

- **Background generation.** Instead of purely random background points, we used climate-stratified sampling. The Köppen-Geiger zone [5] of every presence point was extracted; a large pool of random points over India's land area was generated and assigned zones; background points (3,464, i.e. 1:1) were then sampled from the pool in the same zone proportions as the presences. This forces the classifiers to separate presence from background using finer variation within climate zones, rather than just broad climate type.
- **Splitting and scaling.** A stratified 70/30 train-test split was made (random_state = 42). StandardScaler was fitted on the training data only and then applied to the test data, which avoids leakage from the test set.
- **PCA.** PCA was fitted on the scaled training data; the scree plot and the 90% cumulative-variance rule gave 8 principal components, onto which both training and test sets were projected.
- **Model training.** Decision Tree and KNN were tuned by GridSearchCV with 5-fold cross-validation and ROC AUC as the scoring criterion (Table 2).

Table 2. Hyperparameter search and selected configuration

Model	Search grid	Selected
Decision Tree	max_depth {3, 5, 10, 20, None}; min_samples_split {2, 5, 10}; min_samples_leaf {1, 2, 4}	max_depth = 10; min_samples_split = 10; min_samples_leaf = 4
KNN	n_neighbors {3, 5, 7, 9, 11}; weights {uniform, distance}; metric {euclidean, manhattan}	n_neighbors = 11; weights = distance; metric = euclidean

5. Evaluation Methodology and Outcomes

5.1 Reference paper

The paper evaluated models with sensitivity, the F2-measure (which weights recall more than precision) and ROC AUC. Lasso consistently outperformed ridge, reaching an average AUC of 0.7422 for the native range, and it isolated the key climatic indicators [1]. Emphasising sensitivity is sensible for an invasive species, where missing a suitable area is costlier than over-predicting one.

5.2 Our extension

Models were scored on the isolated 30% test set with accuracy, precision, recall, F1 and ROC AUC. Table 3 reports the results.

Table 3. Test-set performance of our models, with the reference paper's AUC for context

Model	ROC AUC	Accuracy	F1-score	Source
PCA + KNN (k = 11)	0.8888	0.8018	0.8080	Ours
PCA + Decision Tree	0.8168	0.7561	0.7561	Ours
Lasso logistic regression (M. micrantha, native range)	0.7422 (average)	not reported here	not reported here	[1]

KNN clearly outperformed the Decision Tree on every metric, with a gap of about 0.07 in AUC. Both models were free of multicollinearity because of the PCA step. We read the KNN advantage as evidence that presence in L. camara is governed by smooth, non-linear combinations of environmental variables that a single tree, which makes axis-aligned cuts in principal-component space, represents less well.

6. Critical Analysis

This section is our own assessment, not a restatement of either source.

6.1 Strengths

- **Appropriate use of regularisation in the reference paper.** Lasso addresses the stated problem directly and returns a short list of climatic indicators, which is exactly what ecologists and managers need. The method is also easy to explain and to audit.

- **A well-motivated background design in our work.** Climate-stratified background sampling is a thoughtful response to the well-known weakness of presence-only data, and the scaler and PCA are fitted on training data only, so preprocessing does not leak test information.
- **Sensible tuning.** Using ROC AUC as the cross-validation score is consistent with how the models are finally judged, and the grids cover meaningful ranges.

6.2 Weaknesses and concerns

- **Pseudo-absences are not true absences.** Both studies treat background points as class 0, so the models estimate how presence differs from the background, not true habitat suitability. AUC is also known to be a weaker guide in presence-background settings [9], and with a 1:1 ratio the class balance is artificial.
- **Random splitting under spatial autocorrelation.** Nearby locations have similar environments, so a random 70/30 split places near-duplicates of training points in the test set. This inflates scores for any model and may specifically favour KNN, which predicts from the closest training points. The KNN-Decision Tree gap should therefore be treated as a hypothesis rather than a proven result. Spatially blocked validation is the recommended remedy [10].
- **Single split and no significance testing.** Our reported metrics come from one test split. Without repeated splits or confidence intervals we cannot say how much of the 0.07 AUC gap is real. Our wording that KNN was "significantly" better is therefore informal.
- **Loss of interpretability through PCA.** This is the main trade-off against the reference paper. Each of our 8 components blends all 24 variables, so we cannot state, for example, how far from a road L. camara stops being likely. The model predicts reasonably well but explains little, which weakens its usefulness for management.
- **Comparisons across studies are not like-for-like.** The 0.7422 and 0.8888 AUC values come from different species, datasets, background schemes and variable sets. The higher number does not show that PCA + KNN is a better method than lasso; our stratified background and added proximity variables may account for part of the difference.
- **Stratification trades one question for another.** Because background points mirror the presences' climate zones, the model is deliberately blinded to broad climate preference. That is useful for learning local drivers, but it means the model should not be used to claim which climates the species can invade.

7. Limitations and Future Work

7.1 Stated by the reference paper

The authors acknowledge that their correlative approach relied only on climatic variables and ignored other environmental and human factors that drive invasion. They suggest Generalised Additive Models (GAMs) with lasso-type regularisation as a way to capture non-linear physiological responses in future work [1].

7.2 Our own limitations and next steps

We addressed the missing human factors by adding infrastructure and water distances, and the missing non-linearity by using KNN and Decision Trees, but PCA cost us interpretability. The next steps follow from Section 6: (i) evaluate feature importance on the original, unscaled variables using tree ensembles such as Random Forest or gradient boosting; (ii) replace random splits with spatial block cross-validation to account for geographic autocorrelation; (iii) repeat the evaluation over many random splits and report variability; and (iv) compare PCA against variable-selection methods (correlation filtering and Variance Inflation Factor) that keep the original variables.

We have since taken these steps in follow-up work on the same L. camara dataset. Comparing correlation filtering, VIF and PCA across six classifiers, the best model was a tuned Random Forest on VIF-selected variables (test ROC AUC 0.9082), ahead of the PCA-based models in this report. A 2-degree spatial block cross-validation gave a mean AUC of 0.825, against 0.914 for ordinary random cross-validation, which

supports the concern raised in Section 6.2 that random splits give optimistic scores. Feature importance from that model ranked distance to urban areas, railways and roads among the strongest predictors, suggesting that the invasion follows human-disturbed corridors.

8. Conclusion

The reference paper is a well-judged example of adapting a standard model to a domain problem: regularised logistic regression gives a parsimonious, interpretable account of climatic drivers, though its modest AUC suggests that climate alone, entered linearly, does not fully explain invasion. Our extension shows that adding human-disturbance variables and non-linear classifiers can raise predictive performance, with PCA + KNN reaching an AUC of 0.8888, but at the cost of interpretability and with evaluation that is probably optimistic because of spatial autocorrelation. The main lesson is that in SDM the choice of background points, validation design and interpretability matter at least as much as the choice of algorithm.

Appendix: Implementation (PCA, Decision Tree and KNN)

```
import pandas as pd, numpy as np
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier

df = pd.read_csv('data.csv')
non_predictors = ['presence', 'koppen_zone', 'decimalLongitude', 'decimalLatitude']
predictor_cols = [c for c in df.columns if c not in non_predictors]
X, y = df[predictor_cols], df['presence']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y)

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

pca_full = PCA().fit(X_train_scaled)
n_components = np.argmax(np.cumsum(pca_full.explained_variance_ratio_) >= 0.90) + 1
pca = PCA(n_components=n_components)
X_train_pca = pca.fit_transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)

dt_params = {'max_depth': [3, 5, 10, 20, None],
             'min_samples_split': [2, 5, 10],
             'min_samples_leaf': [1, 2, 4]}
dt_grid = GridSearchCV(DecisionTreeClassifier(random_state=42), dt_params,
                       cv=5, scoring='roc_auc', n_jobs=-1)
dt_grid.fit(X_train_pca, y_train)

knn_params = {'n_neighbors': [3, 5, 7, 9, 11],
              'weights': ['uniform', 'distance'],
              'metric': ['euclidean', 'manhattan']}
knn_grid = GridSearchCV(KNeighborsClassifier(), knn_params,
                        cv=5, scoring='roc_auc', n_jobs=-1)
knn_grid.fit(X_train_pca, y_train)
```

References

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